CIE 6002 Matrix Analysis Homework 4

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Declaration: I used LLM for enhancing language, explaining definitions, and checking grammar in this homework.

1. Problem 3.2

Prove the following statements:

1. The eigenvalues of a nilpotent matrix are all zero.
2. The eigenvalues of a unitary matrix have unit absolute values.
3. The eigenvalues of a Hermitian matrix are real.
4. The eigenvalues of a skew-Hermitian matrix are imaginary.

1.1) *Proof.* A square matrix A is called nilpotent if there is a positive integer k s.t. $A^k = 0$.

From the definition of eigenvalues,

$$Ax = \lambda x \tag{1}$$

for some nonzero vector x. Then we have

$$A(Ax) = A(\lambda x) = \lambda(Ax) = \lambda^2 x. \tag{2}$$

Continuing this process, we have

$$A^k x = \lambda^k x. \tag{3}$$

Since $A^k = 0$, we have $\lambda^k x = 0$. Since x is nonzero, we have $\lambda^k = 0$. Therefore, $\lambda = 0$. □

1.2) *Proof.* Unitary matrix A satisfies $A^*A = I$ and $AA^* = I$. Let λ be an eigenvalue of A and x be the corresponding eigenvector. Then we have

$$Ax = \lambda x. \tag{4}$$

Taking conjugate transpose of each side, we have

$$x^*A^* = \overline{\lambda} x^*. \tag{5}$$

Multiplying the Eq. (4) and Eq. (5):

$$x^*A^*Ax = \overline{\lambda}\lambda x^*x \tag{6}$$

$$x^*x = |\lambda|^2 x^*x \tag{7}$$

Since x is nonzero, we have $x^*x > 0$. Therefore, $|\lambda|^2 = 1$ and $|\lambda| = 1$. □

1.3) *Proof.* Let A be a Hermitian matrix and λ be an eigenvalue of A with corresponding eigenvector x. Then we have

$$Ax = \lambda x. \tag{8}$$

Multiplying both sides by x^*, we have

$$x^*Ax = \lambda x^*x = x^*\lambda x. \tag{9}$$

Since x is nonzero, we have

$$x^*A = \lambda x^*. \tag{10}$$

Taking conjugate transpose of each side of Eq. (8), we have

$$x^*A^* = \overline{\lambda}x^*. \tag{11}$$

Since A is Hermitian, we can replace A^* with A in Eq. (11) and get

$$x^*A = \overline{\lambda}x^*. \tag{12}$$

Combining Eq. (10) and Eq. (12), we have $\lambda = \overline{\lambda}$. Therefore, λ is real. □

1.4) *Proof.* The definition of skew-Hermitian matrix A is $A^* = -A$. Let λ be an eigenvalue of A and x be the corresponding eigenvector. Similar to the proof of Part 1.3), we have $x^*A = \lambda x^*$ and $x^*A = -\overline{\lambda}x^*$. Hence, $\lambda = -\overline{\lambda}$. Therefore, λ is imaginary. □

2. Problem 3.5

Using the Schur decomposition theorem to show the following:

1. For $A \in \mathbb{F}^{m\times n}$, there exist unitary matrices U, V, and a positive diagonal matrix Σ such that
$$A = U\Sigma V^*.$$
(This decomposition is called the *singular value decomposition (SVD).*)

2. For each $A \in \mathbb{F}^{n\times n}$, there exist a positive semidefinite matrix P and a unitary matrix U such that
$$A = PU.$$
(This decomposition is called the *polar decomposition.*)

3. A positive definite matrix A can be factorized as $A = T^*T$ where T is upper triangular. (This factorization is called *Cholesky factorization.*)

4. For a $2n \times 2n$ unitary matrix W, there exist $n \times n$ unitary matrices U_1, U_2, V_1, V_2 such that
$$W = \begin{bmatrix} U_1 & 0 \\ 0 & U_2 \end{bmatrix} \begin{bmatrix} C & -S \\ S & C \end{bmatrix} \begin{bmatrix} V_1^* & 0 \\ 0 & V_2^* \end{bmatrix}$$
where
$$C = \begin{bmatrix} \cos\theta_1 & & \\ & \ddots & \\ & & \cos\theta_n \end{bmatrix}, \quad S = \begin{bmatrix} \sin\theta_1 & & \\ & \ddots & \\ & & \sin\theta_n \end{bmatrix}$$
for some $\theta_i \in \left[0, \frac{\pi}{2}\right], i = 1, \ldots, n$. (This decomposition is called *CS decomposition.*)

2.1) Let's consider the matrix A^*A, this matrix is Hermitian because $(A^*A)^* = A^*A$. by the Schur decomposition, there exists a unitary matrix V such that
$$V^*AV = T, \tag{13}$$
where T is upper triangular. Since A^*A is Hermitian, T must be diagonal. Also, A^*A is positive semidefinite, since
$$x^*A^*Ax = (Ax)^*(Ax) \geq 0, \quad \forall x \in \mathbb{F}^n. \tag{14}$$
Since the rank of A^*A doesn't exceed which of A, we can let Σ be the $m \times n$ matrix whose first r diagonal entries are the square roots of the first r diagonal entries of T, and the rest are zero, where r is the rank of A.

For every non-zero σ_i, define a vector $u_i = \frac{1}{\sigma_i}Av_i$, where v_i is the i-th column of V.

Observe that $u_i^*u_j = \frac{1}{\sigma_i\sigma_j}v_i^*A^*Av_j = \frac{1}{\sigma_i\sigma_j}v_i^*(\sigma_j^2 v_j) = \delta_{ij}$. Thus, the set $\{u_i\}$ is orthonormal. Extend this set $\{u_i\}$ to a full orthonormal basis for $\mathbb{F}^m$ to form an $m \times m$ unitary matrix U. By construction, $AV = U\Sigma$. Multiplying both sides by V^* on the right gives:
$$A = U\Sigma V^*$$

2.2) For a given $A \in \mathbb{F}^{n\times n}$, we can apply the SVD to get $A = U\Sigma V^*$. Insertting an identity matrix I between Σ and V^*, we have

$$A = U\Sigma I V^* = (U\Sigma U^*)(UV^*). \tag{15}$$

Here $U\Sigma U^*$ is exactly the Schur decomposition. Since Σ is a positive diagonal matrix, $U\Sigma U^*$ is positive semidefinite. Also, UV^* is unitary since both U and V are unitary. Thus, we have the polar decomposition $A = PU$, where $P = U\Sigma U^*$ and $U = UV^*$.

2.3) For a positive definite matrix A, we can apply the Schur decomposition to get $A = UTU^*$, where U is unitary and T is upper triangular. Since A is positive definite, the diagonal entries of T are positive and T is a diagonal matrix.

Define $B = UT^{1/2}U^*$, then $B^2 = B^*B = A$. By QR decomposition, we can write $B = QT$, where Q is unitary and T is upper triangular. Thus, we have $A = B^*B = T^*Q^*QT = T^*T$.

2.4) Let $W = \begin{bmatrix} W_{11} & W_{12} \\ W_{21} & W_{22} \end{bmatrix}$, where each block is an $n \times n$ matrix.

Because W is unitary ($W^*W = I$ and $WW^* = I$), the block multiplication gives:

(a) $W_{11}^*W_{11} + W_{21}^*W_{21} = I$

(b) $W_{11}W_{11}^* + W_{12}W_{12}^* = I$

(c) $W_{11}^*W_{12} + W_{21}^*W_{22} = 0$

Apply the SVD: $W_{11} = U_1CV_1^*$. Because $W_{11}^*W_{11} \leq I$, its singular values are restricted to the interval $[0, 1]$. Thus, we can define the diagonal matrix $C = \mathrm{diag}(\cos\theta_i)$ for $\theta_i \in [0, \pi/2]$.

Using the first identity, we have

$$W_{21}^*W_{21} = I - W_{11}^*W_{11} = I - V_1C^2V_1^* = V_1(I - C^2)V_1^*. \tag{16}$$

Since $I - C^2 = S^2$ where $S = \mathrm{diag}(\sin\theta_i)$, we have $W_{21}^*W_{21} = V_1S^2V_1^*$. This implies the SVD of W_{21} is $U_2SV_1^*$ for some unitary U_2.

Using the second identity similarly, we have

$$W_{12}W_{12}^* = I - W_{11}W_{11}^* = U_1S^2U_1^*. \tag{17}$$

This means the SVD of W_{12} is $U_1(-S)V_2^*$ for some unitary V_2. Substitute these into the third identity: $(V_1CU_1^*)(U_1(-S)V_2^*) + (V_1SU_2^*)W_{22} = 0$. Simplified as:

$$-V_1CSV_2^* + V_1SU_2^*W_{22} = 0. \tag{18}$$

Multiplying on the left by $S^{-1}V_1^*$ yields $-CV_2^* + U_2^*W_{22} = 0$, meaning $W_{22} = U_2CV_2^*$.Assembling the blocks together:

$$W = \begin{bmatrix} U_1CV_1^* & -U_1SV_2^* \\ U_2SV_1^* & U_2CV_2^* \end{bmatrix} = \begin{bmatrix} U_1 & 0 \\ 0 & U_2 \end{bmatrix} \begin{bmatrix} C & -S \\ S & C \end{bmatrix} \begin{bmatrix} V_1^* & 0 \\ 0 & V_2^* \end{bmatrix}$$

3. Problem 3.14

Consider linear equation

$$Ax = b$$

where $A \in \mathbb{F}^{m\times n}$, $b \in \mathbb{F}^m$, and $x \in \mathbb{F}^n$. If A is square and nonsingular, we can get the unique solution of the equation easily as $x = A^{-1}b$. However we are often interested in the case when A is rectangular or singular.

1. Assume $\operatorname{rank} A = m$. Then there may be infinity many solutions. Show that the minimum norm solution, i.e., the solution with the smallest 2-norm, is $A^*(AA^*)^{-1}b$.
2. Assume $\operatorname{rank} A = n$. Then there may not be solutions. Show that the so-called least square solution, i.e., the x which minimizes $\|Ax - b\|_2$, is $(A^*A)^{-1}A^*b$.
3. In general, we are interested in the minimum norm least square solution, which is the one with the smallest 2-norm among all x that minimizes $\|Ax - b\|_2$. Show that the minimum norm least square solution of the linear equation can be given in the form of $A^\dagger b$ for some matrix $A^\dagger \in \mathbb{F}^{n\times m}$. Here $A^\dagger$ is called the *Moore-Penrose generalized inverse* of A.
4. For linear equation
$$\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} x = \begin{bmatrix} 1 \\ 2 \end{bmatrix},$$
find its minimum norm least square solution.

3.1) Let $x_0 = A^*(AA^*) - 1b$. If there is another solution whose 2-norm is smaller than x_0, we can write it as $x_0 + y$ for some non-zero $y \in \mathbb{F}^n$. We check the inner product of x_0 and y:

$$\langle x_0, y\rangle = y^*A^*(AA^*)^{-1}b = (Ay)^*(AA^*)^{-1}b = 0. \tag{19}$$

Therefore, $\|x_0 + y\|_2^2 = \|x_0\|_2^2 + \|y\|_2^2 > \|x_0\|_2^2$. This contradicts the assumption that $x_0 + y$ has smaller 2-norm than x_0. Thus, x_0 is the minimum norm solution.

3.2) Let $x_0 = (A^*A)^{-1}A^*b$. If there is another solution s.t.

$$\|Ax - b\|_2 < \|Ax_0 - b\|_2, \tag{20}$$

we can write it as $x_0 + y$ for some non-zero $y \in \mathbb{F}^n$. We check the inner product of $A(x_0 + y) - b$ and $Ax_0 - b$:

$$\begin{aligned}
\langle A(x_0 + y) - b, Ax_0 - b\rangle &= (Ax_0 - b)^*(A(x_0 + y) - b) && (21)\\
&= (Ax_0 - b)^*Ay + \|Ax_0 - b\|_2^2 && (22)\\
&= y^*A^*(Ax_0 - b) + \|Ax_0 - b\|_2^2 && (23)\\
&= y^*A^*Ax_0 - y^*A^*b + \|Ax_0 - b\|_2^2 && (24)\\
&= \|Ax_0 - b\|_2^2. && (25)
\end{aligned}$$

Therefore, we have $\langle A(x_0+y)-b, Ax_0-b\rangle = \langle Ax_0-b, Ax_0-b\rangle$, which means $y = 0$. This contradicts the assumption that y is non-zero. Thus, x_0 is the least square solution.

3.3) Let $A = U\Sigma V^*$ be SVD, where U and V are unitary matrices, and Σ is a diagonal matrix containing the r non-zero singular values of A. Minimizing $\|Ax - b\|_2$ is equivalent to minimizing:

$$\|U\Sigma V^* x - b\|_2 = \|\Sigma(V^* x) - U^* b\|_2$$

Let $y = V^* x$ and $c = U^* b$. We want to minimize $\|\Sigma y - c\|_2$. Because Σ only has non-zero diagonal entries $\sigma_1, \ldots, \sigma_r$, the first r components of y must be set to $y_i = c_i/\sigma_i$ (for $i = 1, \ldots, r$) to minimize the norm. For $i > r$, the values of y_i do not affect $\|\Sigma y - c\|_2$, meaning there are infinitely many least square solutions.

To find the *minimum norm* least square solution, we must minimize $\|x\|_2$. Since V is unitary, $\|x\|_2 = \|Vy\|_2 = \|y\|_2$. To minimize $\|y\|_2$, we must set all arbitrary components $y_i = 0$ for $i > r$. This exact construction of y can be written elegantly using the generalized inverse of Σ, denoted $\Sigma^\dagger$, where non-zero elements are inverted and transposed. Thus, $y = \Sigma^\dagger c$. Substituting back x and c:

$$V^* x = \Sigma^\dagger U^* b \implies x = V\Sigma^\dagger U^* b$$

We define the Moore-Penrose generalized inverse as $A^\dagger = V\Sigma^\dagger U^*$. Therefore, the minimum norm least square solution is $x = A^\dagger b$.

3.4) Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Notice that A is a rank-1 matrix since the second row is three times the first row. We can write A as an outer product of two vectors $A = uv^*$ where $u = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $v = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

For a rank-1 matrix $A = uv^*$, its Moore-Penrose pseudoinverse is analytically given by:

$$A^\dagger = \frac{vu^*}{\|u\|_2^2 \|v\|_2^2} \tag{26}$$

Calculate the squared norms:

$$\|u\|_2^2 = 1^2 + 3^2 = 10, \quad \|v\|_2^2 = 1^2 + 2^2 = 5 \tag{27}$$

Now construct $A^\dagger$:

$$A^\dagger = \frac{1}{10 \cdot 5} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 3 \end{bmatrix} = \frac{1}{50} \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \tag{28}$$

The minimum norm least square solution is $x = A^\dagger b$:

$$x = \frac{1}{50} \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \frac{1}{50} \begin{bmatrix} 7 \\ 14 \end{bmatrix} = \begin{bmatrix} 0.14 \\ 0.28 \end{bmatrix} \tag{29}$$

So, the minimum norm least square solution is $x = \begin{bmatrix} 0.14 \\ 0.28 \end{bmatrix}$.

4. Problem 3.22

In numerical computation, say in MATLAB, a subspace $\mathcal{U}$ of $\mathbb{C}^n$ is usually presented by one of its orthonormal bases or equivalently by an isometry U so that $\mathcal{U} = \mathcal{R}(U)$.

1. For two subspaces $\mathcal{U}$ and $\mathcal{V}$ of $\mathbb{C}^n$ represented by U and V respectively, find a way to compute $\mathcal{U} \cap \mathcal{V}$ and $\mathcal{U} + \mathcal{V}$.
2. For $A \in \mathbb{C}^{m\times n}$, find a way to compute $\mathcal{N}(A), \mathcal{R}(A), \mathcal{N}(A)^\perp, \mathcal{R}(A)^\perp$.

4.1) Computing $\mathcal{U} + \mathcal{V}$:
The sum of the two subspaces is spanned by the columns of both U and V. We can form the augmented block matrix $W = [U \quad V]$. To find an isometry representing $\mathcal{R}(W) = \mathcal{U} + \mathcal{V}$, we compute the SVD W:

$$W = P\Sigma Q^* \tag{30}$$

Let $r = \text{rank}(W)$, which corresponds to the number of non-zero singular values in Σ. The isometry for $\mathcal{U} + \mathcal{V}$ is given by P_r, which consists of the first r columns of the unitary matrix P.

Computing $\mathcal{U} \cap \mathcal{V}$:
A numerically stable way to find the intersection is to compute the principal angles between the two subspaces. We form the matrix U^*V and compute its SVD:

$$U^*V = Y\Sigma Z^* \tag{31}$$

The singular values σ_i (the diagonal entries of Σ) represent the cosines of the principal angles between $\mathcal{U}$ and $\mathcal{V}$. The intersection $\mathcal{U} \cap \mathcal{V}$ corresponds to the directions where the angle is 0, meaning $\cos(\theta) = 1$.

Let k be the number of singular values that are equal to 1 (up to a small numerical tolerance). Let Y_k be the matrix consisting of the first k columns of Y corresponding to these singular values $\sigma_i = 1$. The isometry representing the intersection $\mathcal{U} \cap \mathcal{V}$ is given by UY_k. (Equivalently, it can also be computed as VZ_k).

4.2) Compute the full SVD of A:

$$A = \widetilde{U}\Sigma\widetilde{V}^*$$

where $\widetilde{U} \in \mathbb{C}^{m\times m}$ and $\widetilde{V} \in \mathbb{C}^{n\times n}$ are unitary matrices, and $\Sigma \in \mathbb{R}^{m\times n}$ is a diagonal matrix of singular values.

Let $r = \text{rank}(A)$ be the number of non-zero singular values (typically determined by counting singular values greater than a numerical threshold ϵ). We can partition the unitary matrices into blocks based on r:

$$\widetilde{U} = \begin{bmatrix}\widetilde{U}_1 & \widetilde{U}_2\end{bmatrix}, \quad \widetilde{V} = \begin{bmatrix}\widetilde{V}_1 & \widetilde{V}_2\end{bmatrix}$$

where $\widetilde{U}_1 \in \mathbb{C}^{m\times r}$ contains the first r columns of $\widetilde{U}$, $\widetilde{U}_2 \in \mathbb{C}^{m\times(m-r)}$ contains the remaining columns, $\widetilde{V}_1 \in \mathbb{C}^{n\times r}$ contains the first r columns of $\widetilde{V}$, and $\widetilde{V}_2 \in \mathbb{C}^{n\times(n-r)}$ contains the remaining columns.

The isometries representing the four fundamental subspaces are directly given by these block matrices:

- **Range of** A: $\mathcal{R}(A)$ is represented by the isometry $\widetilde{U}_1$.
- **Orthogonal complement of the Null Space**: $\mathcal{N}(A)^\perp = \mathcal{R}(A^*)$ is represented by the isometry $\widetilde{V}_1$.
- **Null Space of** A: $\mathcal{N}(A)$ is represented by the isometry $\widetilde{V}_2$.
- **Orthogonal complement of the Range**: $\mathcal{R}(A)^\perp = \mathcal{N}(A^*)$ is represented by the isometry $\widetilde{U}_2$.

5. Problem 4.2

Let $A, B \in \mathcal{H}(n, \mathbb{F})$. Prove the following statements:

1. If $A > 0$, then there exists $T \in \mathcal{GL}(n, \mathbb{F})$ such that T^*AT and T^*BT are simultaneously diagonal.
2. If $A \geq B > 0$, then $A^{-1} \leq B^{-1}$.
3. If $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \geq B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} > 0$ where the partitions of A and B are consistent, then $A/_{11} \geq B/_{11}$.

5.1) Since $A > 0$, there exists a unique positive definite square root $A^{1/2}$, and its inverse $A^{-1/2}$ is also positive definite. Let $P = A^{-1/2}$. We have $P^*AP = A^{-1/2}AA^{-1/2} = I$.

Consider the matrix P^*BP. Because B is Hermitian and P is Hermitian, P^*BP is also Hermitian. Thus, there exists a unitary matrix U and a real diagonal matrix D such that:

$$U^*(P^*BP)U = D$$

Let $T = PU$. Since P is non-singular and U is unitary (hence non-singular), $T \in \mathcal{GL}(n, \mathbb{F})$. Now, let us evaluate T^*AT and T^*BT:

$$T^*AT = (PU)^*A(PU) = U^*(P^*AP)U = U^*IU = I$$

$$T^*BT = (PU)^*B(PU) = U^*(P^*BP)U = D$$

Both I and D are diagonal matrices. Therefore, the matrix T simultaneously diagonalizes A and B.

5.2) Since $A \geq B > 0$, both A and B are positive definite, so their inverses and square roots exist. The condition $A \geq B$ implies $A - B \geq 0$.

Multiply both sides of $A \geq B$ by $B^{-1/2}$ on the left and right. Since congruence transformations preserve the Loewner partial ordering, we have:

$$B^{-1/2}AB^{-1/2} \geq B^{-1/2}BB^{-1/2} = I$$

Let $C = B^{-1/2}AB^{-1/2}$. Since $C \geq I$, all eigenvalues of C are greater than or equal to 1, and all eigenvalues of C^{-1} must be between 0 and 1. This implies:

$$C^{-1} \leq I$$

Substitute the definition of C back into the inequality:

$$(B^{-1/2}AB^{-1/2})^{-1} = B^{1/2}A^{-1}B^{1/2} \leq I$$

Multiply on the left and right by $B^{-1/2}$ to isolate A^{-1}:

$$B^{-1/2}(B^{1/2}A^{-1}B^{1/2})B^{-1/2} \leq B^{-1/2}IB^{-1/2}$$

$$A^{-1} \leq B^{-1}$$

5.3) The notation $A/_{11}$ denotes the Schur complement of block A_{11} in A, defined as $A/_{11}= A_{22}-A_{21}A_{11}^{-1}A_{12}$.

From the block matrix inversion formula. For a positive definite block matrix $X = \begin{bmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{bmatrix}$, the bottom-right block of its inverse X^{-1} is precisely $(X/_{11})^{-1}$.

From part 5.2), we know that if $A \geq B > 0$, then $B^{-1} \geq A^{-1} > 0$. If a matrix is positive semi-definite, all its principal submatrices must also be positive semi-definite. Therefore, the inequality holds for the bottom-right blocks of B^{-1} and A^{-1}:

$$(B^{-1})_{22} \geq (A^{-1})_{22} > 0$$

Using the property of the block matrix inverse, we substitute the bottom-right blocks with the inverses of their respective Schur complements:

$$(B/_{11})^{-1} \geq (A/_{11})^{-1} > 0$$

Since Schur complements of positive definite matrices are themselves positive definite, $(A/_{11})^{-1}$ and $(B/_{11})^{-1}$ are positive definite. We can apply the result of part 5.2) one more time. Taking the inverse of both sides reverses the inequality:

$$((B/_{11})^{-1})^{-1} \leq ((A/_{11})^{-1})^{-1}$$

$$B/_{11} \leq A/_{11}$$

which means $A/_{11} \geq B/_{11}$.